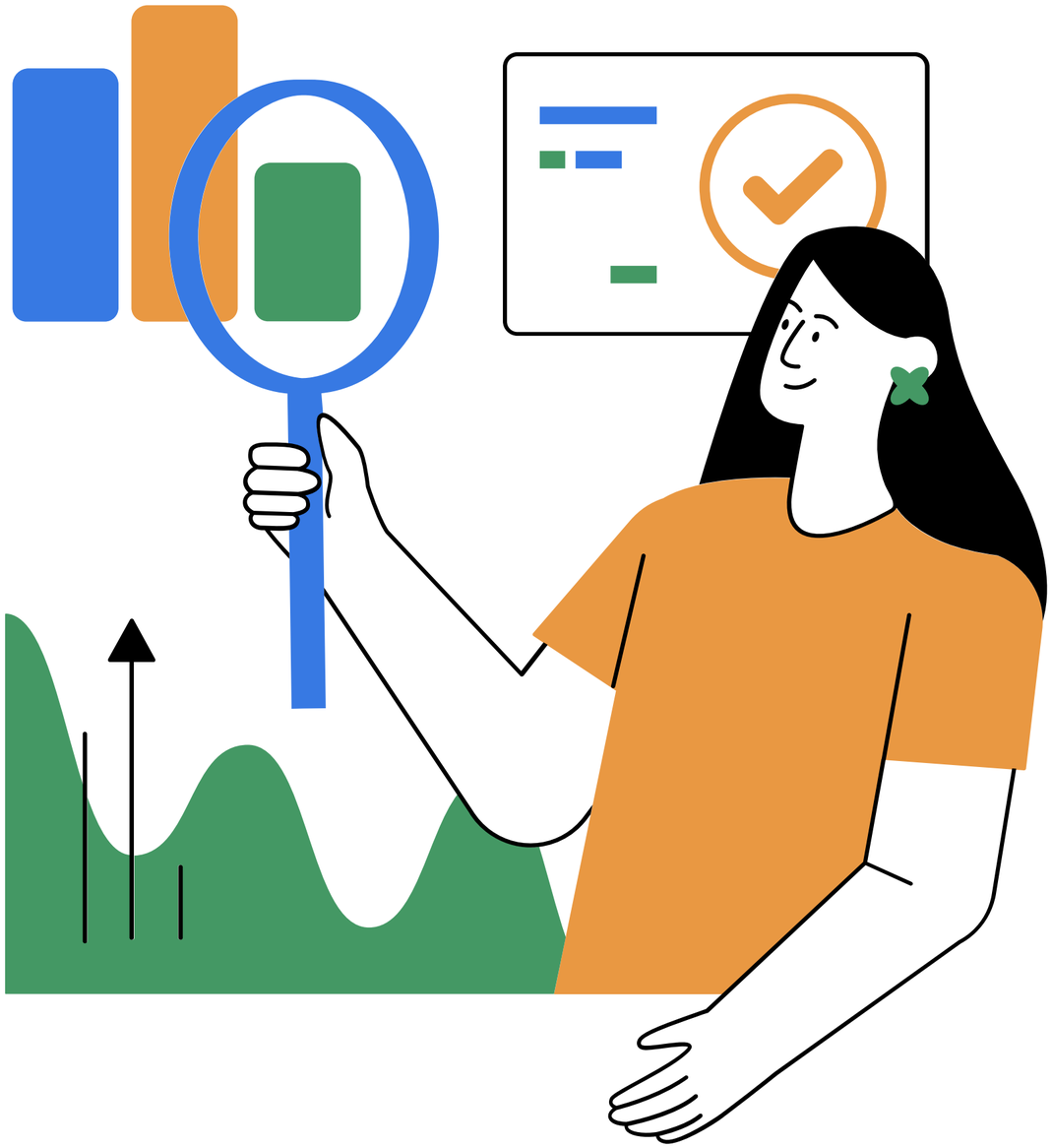
BSAN 360

Factors Influencing Product Sales on Amazon

by Simar Kaur



Dataset

- Dataset contains **50,000** transaction-level observations
- Covers e-commerce purchases from 2022 to 2023
- Each row represents a single customer order
- Includes 6 product categories:
  - Beauty, Books, Fashion, Home & Kitchen, Electronics, Sports
- Covers 4 geographic regions:
  - Asia, Europe, Middle East, North America
- Key variables used in analysis:
  - Total revenue (target variable)
  - Quantity sold
  - Price and discount percentage
  - Customer rating (1–5 scale)
  - Review count
- Dataset is complete with no missing values or duplicates, ensuring reliable analysis

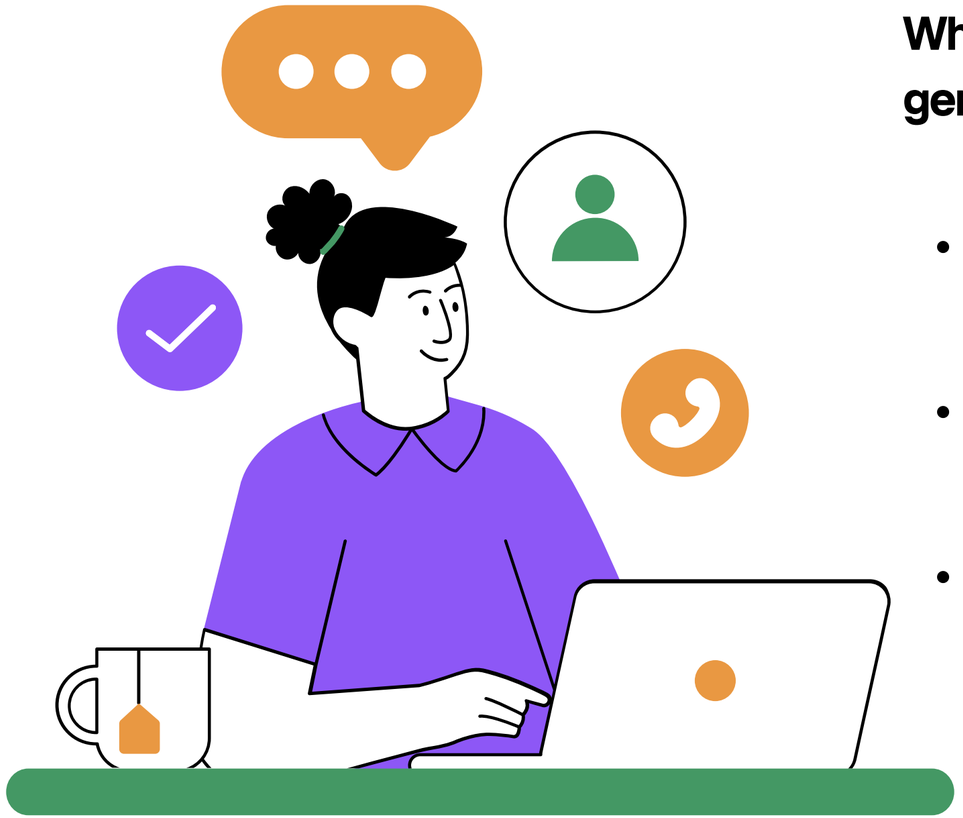
Descriptive Statistics

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Price ($)	252.51	143.03	5.01	127.84	252.97	376.34	499.99
Discount (%)	13.34	9.85	0	5	10	20	30
Quantity Sold	3	1.42	1	2	3	4	5
Rating	3	1.15	1	2	3	4	5
Review Count	249.33	144.25	0	125	250	374	499
Total Revenue ($)	657.33	526.22	3.84	240.32	505.41	968.97	2499.55

Data Preparation

- Standardized column names for consistency
- Converted order_date into datetime format for time-based analysis
- Verified dataset quality:
  - No duplicate observations
  - No missing values across all variables
- Cleaned categorical variables:
  - Standardized payment method labels
- Created new variables:
  - Order month and year for temporal analysis
- Final dataset:
  - **50,000 observations and 13 variables**
  - Focused on variables relevant to sales performance and customer behavior

Research Questions



Which product categories generate the highest revenue?

- Compare total revenue and quantity sold across categories
- Evaluate whether certain categories consistently outperform others
- Determine if differences in performance are meaningful or relatively small

Do ratings and review counts influence sales performance?

- Analyze relationship between customer ratings and revenue outcomes
- Examine whether products with more reviews generate higher sales
- Assess whether customer feedback metrics are strong predictors of revenue

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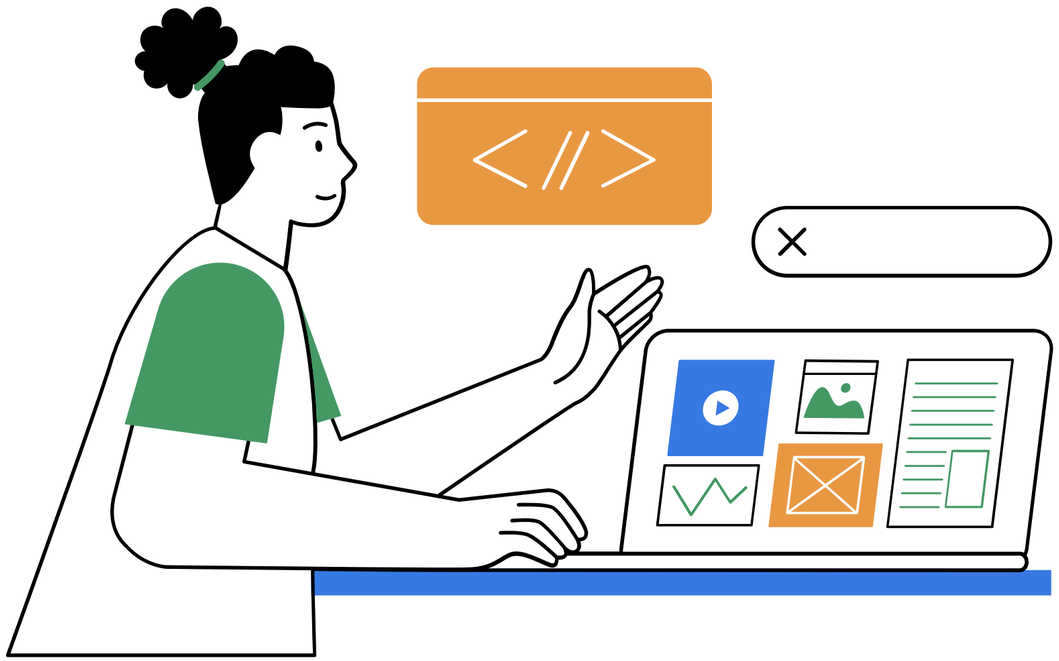


Methodology

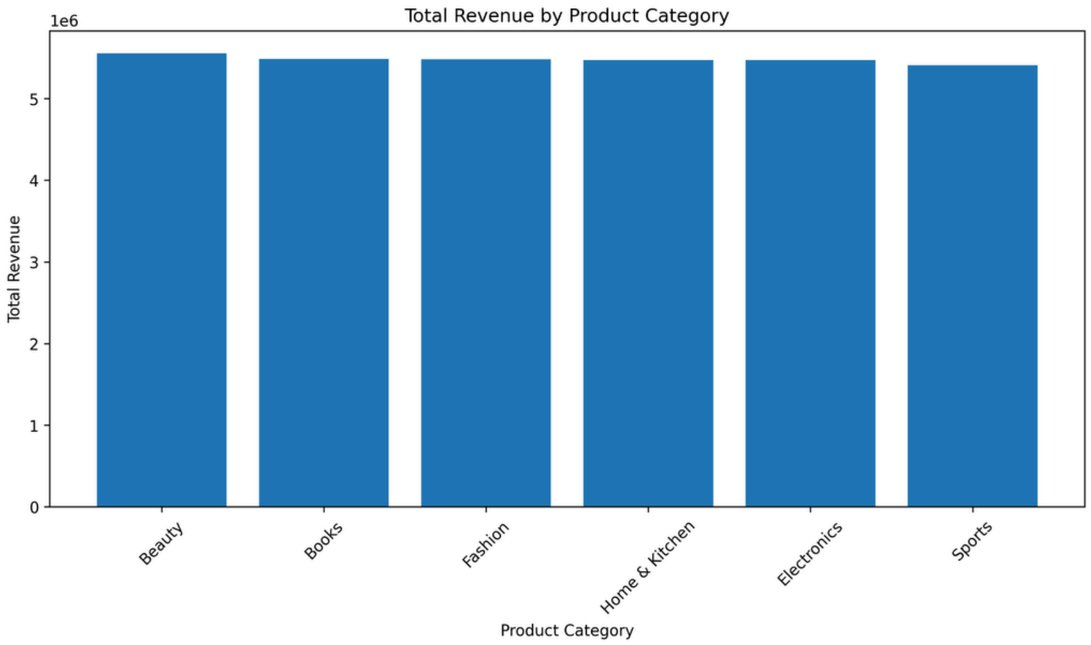
- **Category Performance Analysis:**
  - Grouped data by product category
  - Calculated:
    - Total revenue
    - Total quantity sold
    - Average price and customer rating
  - Compared performance across categories
- **Customer Feedback Analysis:**
  - Examined relationship between:
    - Ratings and total revenue
    - Review count and total revenue
  - Used correlation measures and visual analysis
- **Visualization Approach:**
  - Bar charts to compare category performance
  - Scatter plots to evaluate relationships between variables
  - Trend analysis using review count

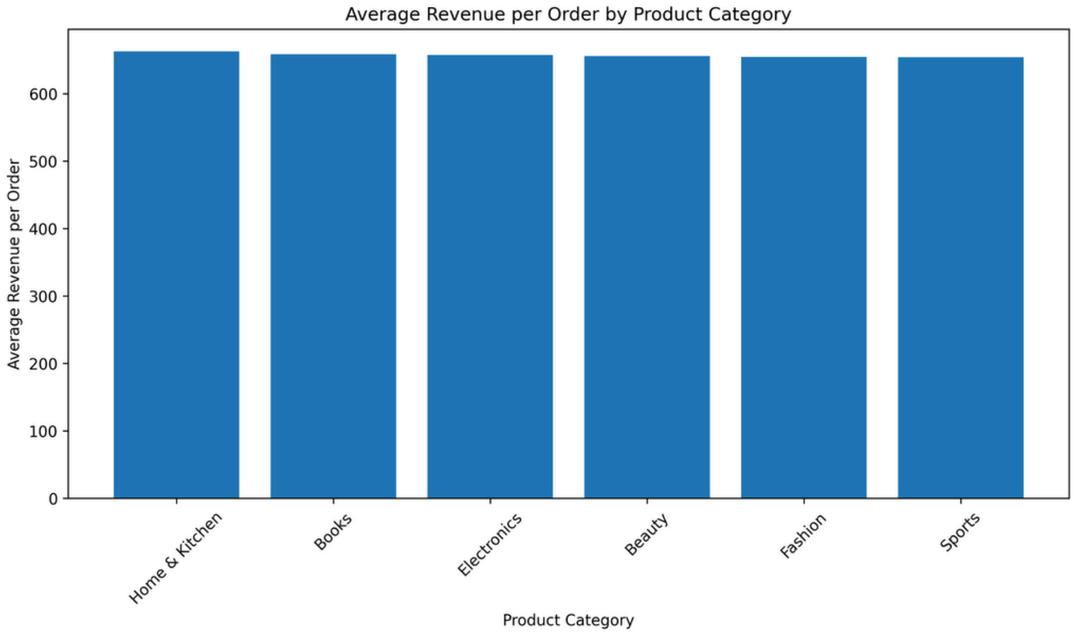
Category Performance

- Total revenue is very similar across all product categories, ranging from:
  - $5.41M (Sports) to $5.55M (Beauty)
- The difference between the highest and lowest category is only ~$140K, which is small relative to total sales
- Total quantity sold is also consistent:
  - Approximately 24,700 to 25,400 units across all categories

- Average revenue per order is nearly identical across categories:
  - Roughly $650–$670 per order
- Average prices are also highly consistent (~$252 across categories)

- **Key Insights:**
  - **No category significantly outperforms others in revenue or volume**
  - **Differences in total revenue are driven by small variations in quantity sold, not pricing**
  - **Product category is not a strong differentiator of sales performance**
  - **Variation across categories is small (~3%) and not economically meaningful**



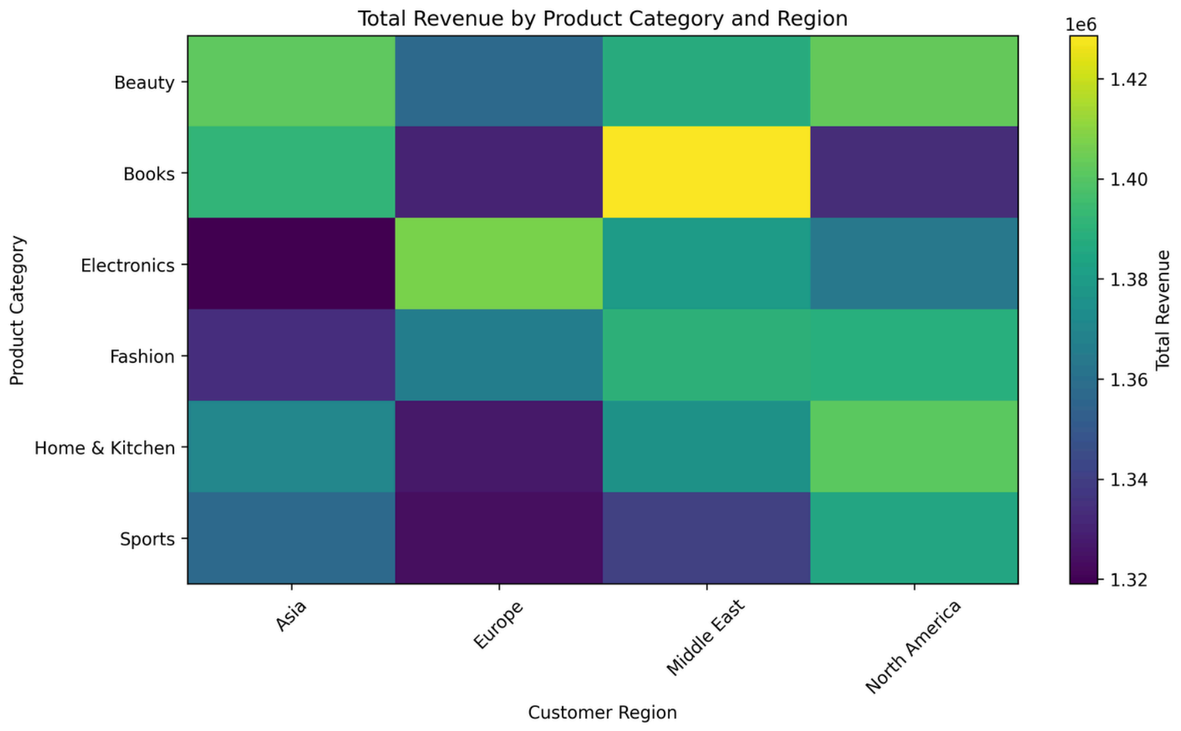


Revenue by Category and Region

- The heatmap shows that revenue levels are relatively similar across all category–region combinations, with no consistent pattern indicating that either product category or region drives differences in sales

- **Some localized variation is present:**
  - Books generate slightly higher revenue in the Middle East compared to other regions
  - Electronics perform relatively better in Europe but show weaker performance in Asia
  - North America appears consistent across most categories without extreme highs or lows’

- However, these differences are small and not consistent across categories:
  - No single region consistently produces the highest revenue
  - No category dominates across all regions

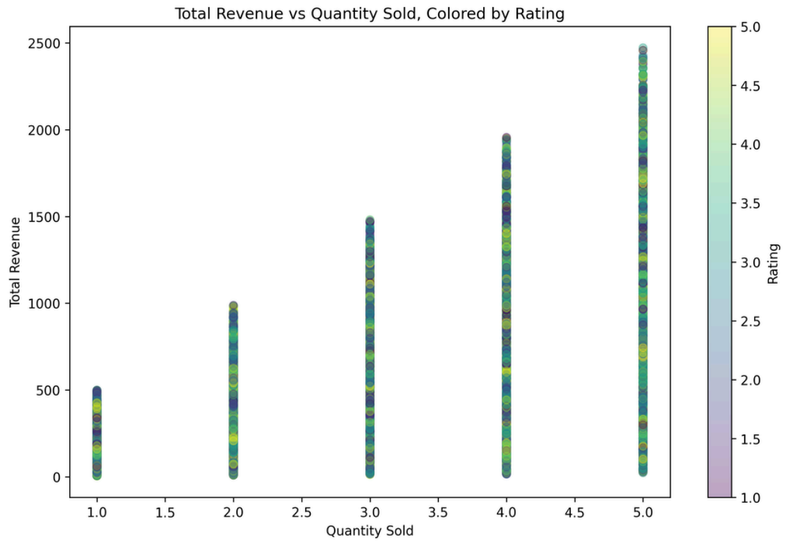


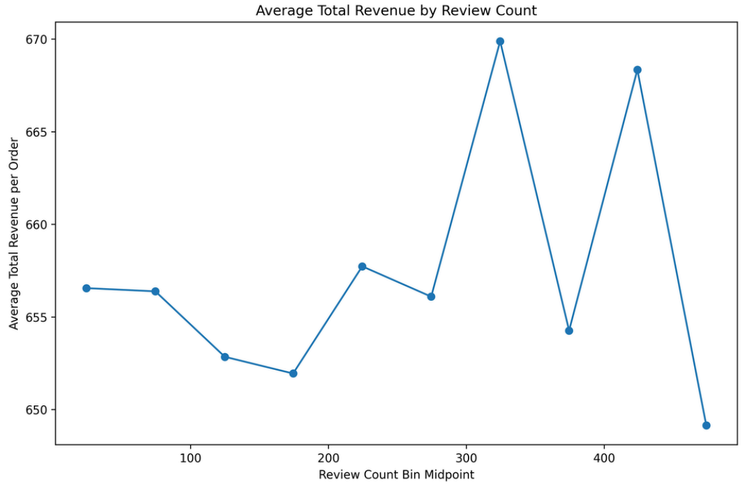
Ratings and Sales

- Strong positive relationship between:
  - Quantity sold and total revenue (expected)
- Ratings are distributed evenly across all levels of revenue
- High-rated and low-rated products appear across both low and high revenue
- No visible trend — ratings appear randomly distributed across all revenue levels

Insight:

- **Customer ratings do not significantly influence revenue outcomes**
- **Sales are driven more by transaction volume than perceived product quality**
- **Ratings provide no predictive signal for sales performance**





Do Ratings and Reviews Influence Sales?

What the results show

- All correlations are extremely close to zero, indicating no meaningful relationship
- → Ratings and reviews move independently from sales
- Regression results confirm this:
  - Very small coefficients for both variables
  - $R^2 = 0.0$, meaning the model explains none of the variation in sales

What this means

- Products with higher ratings or more reviews do not consistently generate higher revenue
- Customer feedback varies, but sales remain widely spread regardless of those values

Interpretation

- Ratings and reviews may reflect customer experience, but they are not strong drivers of purchasing behavior in this dataset
- Sales outcomes are likely driven more by:
  - Demand (quantity sold)
  - Pricing and product availability

Variable	Revenue Model Coefficient	Quantity Model Coefficient
Intercept	652.85	3.01
Rating	0.83	-0.0035
Review Count	0.0081	0.000007
R^2	0	0

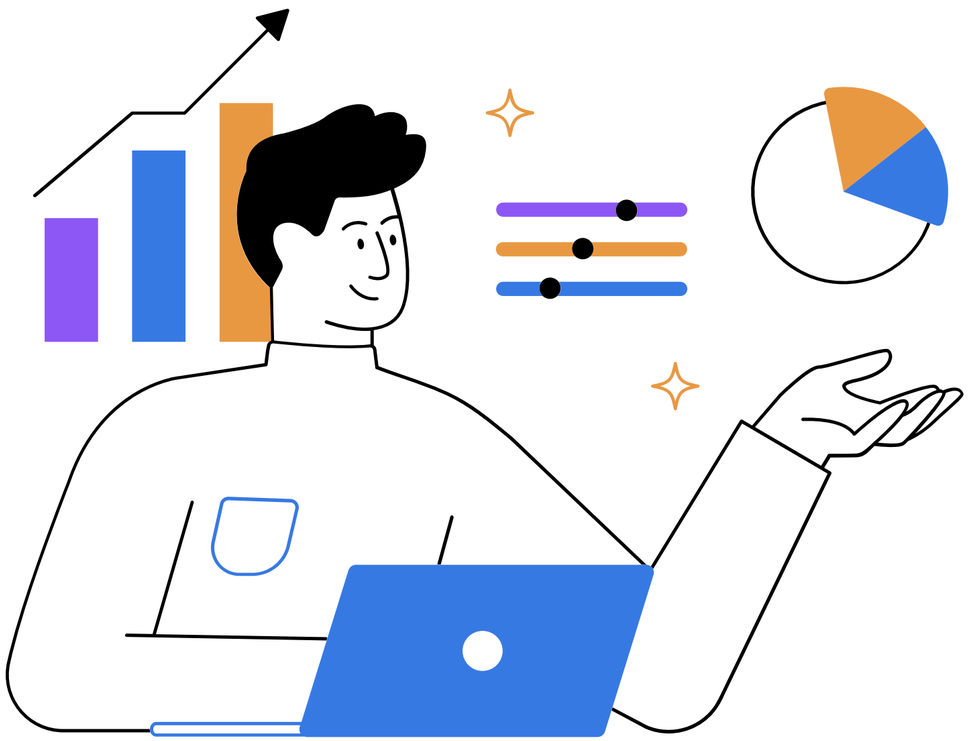
Key Insights

I initially expected ratings and review counts to be strong drivers of sales, based on typical e-commerce assumptions

- However, the analysis shows no meaningful relationship between customer feedback and revenue
  - Both correlation and regression results are effectively zero
    - Higher-rated or more-reviewed products are not consistently performing better
- At the same time, product categories also show very similar performance, with only small differences in total revenue

Main Insight:

- Both category and customer feedback fail to explain differences in sales, suggesting that revenue is driven by other factors not captured in this analysis, rather than category or customer feedback



Implications for business/society

Managerial Implications (Business)

- Companies should not rely on product category or customer ratings to drive revenue growth
- Revenue is primarily driven by demand (quantity sold) rather than perceived product quality
- Firms should prioritize:
  - *Pricing strategy*
  - *Promotions and visibility*
  - *Inventory and product availability*
- Investing heavily in improving ratings or expanding categories is unlikely to generate meaningful returns

Broader Implications (Business / Society)

- Consumer feedback metrics (ratings, reviews) may reflect experience but are not strong predictors of purchasing behavior
- Businesses should be cautious about over-relying on surface-level metrics when making strategic decisions
- Data-driven decision-making requires identifying true drivers of demand, not just commonly tracked indicators